AI Queue Manager

MOHAMED ASMATHULLA G[1]

I. PROJECT OBJECTIVE

Modify an existing application into an **AI-driven automation system** (using n8n or any free agent) to manage token-based user queues across multiple industries (hospitals, schools, banks, hotels, etc.) while preventing duplication.

II. USER ROLES AND ACCESS CONTROL

Role	Description
Main Admin	Super administrator who manages all industry admin requests (accept/reject with email/message), sends default passwords & terms, and controls the entire application.
Industry Admin	Registers for approval. Once accepted, can log in, reset password, create branches, configure dashboards, and manage operators/service providers.
Queue Operator	Verifies user details, generates unique tokens, allocates service providers, and sees queue counts with user suggestions.
Service Provider	Role-based (doctor, teacher, manager). Views user details, uses AI suggestions, manages transactions, cash prizes, and checkboxes.
End User	Registers/logs in, selects place/industry, fills details, receives token, views queue status & suggestions.

III. REGISTRATION AND LOGIN RULES

- **Main Admin** login is **not displayed** on login page.
- Separate **register page** for industry admins, operators, and service providers.
- Industry admins see **"waiting"** until Main Admin approves or rejects.
- Operators/Providers register using a **unique branch code** generated by the industry admin.
- **Forgot password** sends a default password (all roles except operators/providers who do not get default passwords).

IV. BRANCH CREATION (Checkbox Method Only)

Each branch includes:

- Branch name, details, type (hospital, school, hotel, bank, or other)
- Example format: Hospital → Suggestion: Consultation:300, Service Charge:100 | Fields: name:text, phone:text, need:text
- Three dashboard sites: **User Site**, **Queue Operator Site**, **Service Provider Site**

V. DASHBOARD CONFIGURATIONS

Site	Configurable Items (Checkboxes)
Service Provider	Display user details, previous/next/current user, suggestion text boxes (custom count & titles), search checkboxes (price, name), cash prize (total, average, count, aggregations), transactions (send/received/pending/failure), AI suggestion box (generated on button click)
Queue Operator	Same as above + **edit** or **read** permission per item. Can allocate service providers to users.
User	Key-value attributes (int, text, float, image). Checkboxes for: preview suggestions, current queue count, generate/reject token, cash prize, transactions, operator/provider names/contacts, total queue count

VI. TOKEN AND QUEUE MANAGEMENT

- Token validity: **30–60 minutes**
- Notifications sent **every 10 minutes before expiry**
- If user not entered within time → token **auto-rejected** + notification
- Missing/pending users auto-reordered after next customer
- Queue operator verifies with **"Customer In"** click
- **AI agent** auto-detects duplicate tokens

VII. DATA SHARING OPTION

When creating branches, industry admin selects:

- **Allow all branches** – share user details across all branches
- **Allow selected branch only** – restrict sharing

VIII. TECHNICAL AND DEVOPS REQUIREMENTS

Component	Specification
Style	Classical, real-time application
Database	Normalized (future cloud DB support)
Containerization	Docker
Orchestration	Kubernetes with **2 replicasets**
AI Automation	n8n or free agent for suggestions & duplicate detection
DevOps Tools	CI/CD, monitoring, logging as needed

IX. SUMMARY

Build a centralized, AI-powered queue management system with role-based access, multi-branch support, customizable dashboards (using only checkboxes, no typing), unique token validation, auto-reordering, and full DevOps deployment (Docker + K8s + cloud-ready DB).